Molien's Theorem to compute Invariants

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Declaration

I, Soham Sarkar hereby declare that the matter of this project report entitled *Molien's Theorem to compute Invariants* is record of the research work carried out by me, at the Department of Mathematics, Savitribai Phule Pune Univerity, Pune under the supervision of Prof. Ganesh S. Kadu, Department of Mathematics, Savitribai Phule Pune University in partially fulfilling the requirements of the course MT-34 : Algebraic Geometry

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Abstract

In this project, we are discussing the finite generation problem of rings of invariants on finite dimensional Hilbert spaces leading up to formulation of *Molien's series*. In the first chapter, we have discussed some preliminary concepts and some prerequisites for unitary representation of groups on symmetric algebras. In the second chapter, we proved the induction of unitary property on graded algebras and discussed some properties of Reynold's operators. In the third chapter, we have explored the spectrum of these induced unitary operators and introduced the algebra of invariants. Finally, in the last chapter, we focus on the constructing the series using the notion of traces and also provide an example.

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Chapter 1

Introduction

In Invariant theory, when we consider a polynomial in $K[x_1, x_2, ..., x_n]$ (for an algebraically closed field K) to be invariant under finite matrix group G $\subset$ GL(n,K), one of the fundamental question which arise about the ring of invariants $K[x_1, x_2, ..., x_n]^G$, is as follows :
Finite Generation problem : Can we find finitely many homogeneous invariants $f_1, f_2, ..., f_m$, such that every invariant in $K[x_1, x_2, ..., x_n]$ is a polynomial in $f_1, f_2, ..., f_m$?

Emmy Noether was able to prove a wonderful theorem which stated that we can always find finitely many of these invariants that generate $K[x_1, x_2, ..., x_n]^G$ by the notion of Reynold's operator of G on monomials whose bound on the total degree is $\leq$ |G|.

However, for large matrix groups, the computations would go worse with more variables getting involved. An efficient method for finding a generating set of invariants is using *Molien's series*, which enables us to predict in advance the number of linearly independent homogeneous invariants of a given total degree.

A look at the formula $\phi_G(\lambda) = \frac{1}{|G|} \sum_{g \in G} \frac{1}{det(id - \lambda T_g)}$ suggests that the elements of linear algebra and representation theory are linked together.

Preliminaries :

Consider a finite dimensional Hilbert space V ($\simeq \mathbb{C}^n$) with an orthonormal basis $B = (e_1, e_2, ..., e_n)$ and x = $(x_1, x_2, ..., x_n)$ be an orthonormal basis of it's corresponding dual space V*, such that $x_i(e_j) = \delta_{ij}$, $\forall$ $1 \leq i, j \leq$ n. Consider a group G, such that $\forall$ g $\in$ G, the map V $\rightarrow$ V mapping v $\mapsto$ g$\cdot$v is a unitary bijective linear map. So, the corresponding matrices $[\mathrm{g}\cdot\mathrm{v}]_B = [\mathrm{g}]_{B,B}\cdot[v]_B$, where $[\mathrm{g}]_{B,B}$ is unitary.

The left action as stated above is a unitary representation of G i.e. a G-module. Considering $x_1, x_2, ..., x_n$ as variables, the symmetric algebra on V^* i.e. $\mathrm{Sym}(\mathrm{V}^*)$ is $\mathrm{R} := \mathbb{C}[x_1, x_2, ..., x_n]$ of polynomial functions V $\rightarrow \mathbb{C}$ i.e. complex linear forms on V. It is naturally graded by degree as $\mathrm{R} = \oplus_{d\in\mathbb{N}}\mathrm{R}_d$, where R_d is the vector space spanned by the polynomials of total degree d. In particular, we have $\mathrm{R}_0 = \mathbb{C}$ and $\mathrm{R}_1 = \mathrm{V}^*$.

Proposition 1.1: Let V, G, R be as stated above. Then, the map $\cdot$: $\mathrm{G}\times\mathrm{R} \rightarrow \mathrm{R}$ mapping $(\mathrm{g},\mathrm{f}) \mapsto \mathrm{g}\cdot\mathrm{f}$ defined by $(\mathrm{g}\cdot\mathrm{f})(\mathrm{v}) := \mathrm{f}(\mathrm{g}^{-1}\cdot\mathrm{v})$ for $\mathrm{v} \in \mathrm{V}$ is a left action i.e. the action of G on V can be lifted to an action of G on R.

Proof : For $\mathrm{v} \in \mathrm{V}$, g, $\mathrm{h} \in \mathrm{G}$ and $\mathrm{f} \in \mathrm{R}$, we check that
i) $((\mathrm{hg}).\mathrm{f})(\mathrm{v}) = \mathrm{f}((\mathrm{hg})^{-1}.\mathrm{v}) = \mathrm{f}((\mathrm{g}^{-1}\mathrm{h}^{-1}).\mathrm{v}) = \mathrm{f}(\mathrm{g}^{-1}.(\mathrm{h}^{-1}.\mathrm{v}))$
$= (\mathrm{g}.\mathrm{f})(\mathrm{h}^{-1}.\mathrm{v}) = (\mathrm{h}.(\mathrm{g}.\mathrm{f}))(\mathrm{v})$
ii) $(1.\mathrm{f})(\mathrm{v}) = \mathrm{f}(1^{-1}.\mathrm{v}) = \mathrm{f}(1.\mathrm{v}) = \mathrm{f}(\mathrm{v})$ and

Proposition 1.2: Let V, G, R be stated as above. $\forall$ $\mathrm{g} \in \mathrm{G}$, the map $\mathrm{T}_g : \mathrm{R} \rightarrow \mathrm{R}$, mapping $\mathrm{f} \mapsto \mathrm{g}.\mathrm{f}$ is an algebra automorphism, which preserves the grading i.e. $\mathrm{g}.\mathrm{R}_d \subset \mathrm{R}_d$.

Proof : For $\mathrm{v} \in \mathrm{V}$, $\mathrm{g} \in \mathrm{G}$, $\mathrm{c} \in \mathbb{C}$ and f, $\mathrm{f}' \in \mathrm{R}$, we check that
1. $(\mathrm{g}.(\mathrm{f} + \mathrm{f}'))(\mathrm{v}) = (\mathrm{f} + \mathrm{f}')(g^{-1}.\mathrm{v}) = \mathrm{f}(g^{-1}.\mathrm{v}) + \mathrm{f}'(g^{-1}.\mathrm{v})$
   $= (\mathrm{g}.\mathrm{f})(\mathrm{v}) + (\mathrm{g}.\mathrm{f}')(\mathrm{v}) = (\mathrm{g}.\mathrm{f} + \mathrm{g}.\mathrm{f}')(\mathrm{v})$,
   thus $\mathrm{g}.(\mathrm{f} + \mathrm{f}') = \mathrm{g}.\mathrm{f} + \mathrm{g}.\mathrm{f}'$
2. $(\mathrm{g}.(\mathrm{f}.\mathrm{f}'))(\mathrm{v}) = (\mathrm{f}.\mathrm{f}')(g^{-1}.\mathrm{v}) = \mathrm{f}(g^{-1}.\mathrm{v}).\mathrm{f}'(g^{-1}.\mathrm{v}) = (\mathrm{g}.\mathrm{f})(\mathrm{v}).(\mathrm{g}.\mathrm{f}')(\mathrm{v})$
   $= (\mathrm{g}.\mathrm{f}.\mathrm{g}.\mathrm{f}')(\mathrm{v})$,
   thus $\mathrm{g}.(\mathrm{f}.\mathrm{f}') = \mathrm{g}.\mathrm{f}.\mathrm{g}.\mathrm{f}'$
3. $(\mathrm{g}.(\mathrm{cf}))(\mathrm{v}) = (\mathrm{cf})(g^{-1}.\mathrm{v}) = \mathrm{c}(\mathrm{f}(g^{-1}.\mathrm{v})) = \mathrm{c}((\mathrm{g}.\mathrm{f})(\mathrm{v})) = (\mathrm{c}(\mathrm{g}.\mathrm{f}))(\mathrm{v})$
4. By part 2, it is clear that the grading is preserved.
5. To show that $\mathrm{f} \mapsto \mathrm{g}.\mathrm{f}$ is bijective, it is enough to show that this mapping is injective on the finite dimensional homogeneous components R_d. Define this map, say $\mathrm{T}_g^d : \mathrm{R}_d \mapsto \mathrm{R}_d$, mapping $\mathrm{f} \mapsto \mathrm{g}.\mathrm{f}$. Now, if $\mathrm{f} \in \ker(\mathrm{T}_g^d)$, then $\mathrm{g}.\mathrm{f} = 0 \in \mathrm{R}_d$, i.e. g.f is a polynomial mapping from V to $\mathbb{C}$ of degree d vanishing identically, $\forall$ $\mathrm{v} \in \mathrm{V}$ i.e. $(\mathrm{g}.\mathrm{f})(\mathrm{v}) = 0$. By definition of the extended action on R, we have $\forall$ $\mathrm{v} \in \mathrm{V}$: $\mathrm{f}(g^{-1}.\mathrm{v}) = 0$. Since G acts on V, this

implies that $\forall$ v $\in$ V: f(v) = 0 i.e. f is the zero mapping. Since the base field $\mathbb{C}$ is of characteristic 0, this implies that f is the zero polynomial, which we may view as an element of every R_d.

6. Note that every T_g^d is surjective, since all group elements have their inverse in G.

These two propositions together gives a homomorphism from G into Aut(R). As the monomials $x_1, x_2, ..., x_n$ of degree 1 form a basis for R_1, it follows from the proposition that in general, the monomials of total degree d in the linear forms form a basis x_d of R_d.

For f $\in R_d$ and v $\in$ V , define
$T_g^d : R_d \to R_d$, f $\mapsto$ g.f : $R_d \to \mathbb{C}$, v $\mapsto$ f($g^{-1}\cdot$v) = f($T_{g^{-1}}$v).
If f, f′ are eigen vectors of T_g^1 corresponding to eigen values λ, λ', then, f.f′ is an eigen vector of T_g^1 with eigen value $\lambda.\lambda'$ because
$T_g^1(f.f') = T_g^1(f).T_g^1(f') = (\lambda f).(\lambda f') = (\lambda\lambda')(f.f')$

We end this section with two little facts needed in the next section.

Proposition 1.3: For g $\in$ G, $T^1_{g^{-1}} = (T^1_g)^{-1}$.

Proof : Since $\dim(V^*) < \infty$, it is sufficient to prove that
$T^1_{g^{-1}} \circ T^1_g = id_{V^*}$. For arbitrary f $\in V^*$, we see that
$(T^1_{g^{-1}} \circ T^1_g)(f) = T^1_{g^{-1}}(T^1_g(f)) = T_{g^{-1}}(g.f) = g^{-1}.(g.f) = (g^{-1}g).f = f$.

Proposition 1.4: For g $\in$ G and f $\in V^*$, $T^1_g(f) = f \circ T_{g^{-1}}$.

Proof : For v $\in$ V, we see that
$(T^1_g(f))(v) = (g.f)(v) = f(g^{-1}.v) = f(T_{g^{-1}}(v))$.

Chapter 2

Induced Unitary Operator

We require a unitary representation of G, that is the operators T_g:V $\mapsto$ V need to be unitary i.e. $\forall$ g $\in$ G, $(T_g)^{-1} = (T_g)^*$. The first goal of this section is to show that this implies that the induced operators T_g^d: $R_d \to R_d$ mapping f $\mapsto$ g.f are also unitary. In order to understand the operator duals of V and V^*, we need to look at their inner products first. So, we may assume that the operators T_g are unitary with respect to standard inner product $\langle u, v\rangle = [u]_{B,B}.\overline{[v]_{B,B}}$, the dot product.

Before we can speak of unitarity of the induced operators T_g^d, we have to make clear which inner product applies on $\mathbb{R}^1 = V^*$.

Consider the following diagrams :

$$\begin{tikzcd}
V^* \arrow[r, "T_g^1"] \arrow[d, "P"'] & V^* \arrow[d, "P"] \\
V \arrow[r, "T_g"] & V
\end{tikzcd}$$

and

$$\begin{tikzcd}
V^* \arrow[r, "T_g^{\rtimes}"] \arrow[d, "P"'] & V^* \arrow[d, "P"] \\
V \arrow[r, "T_g^*"] & V
\end{tikzcd}$$

where $V^* \equiv R^1$, P denotes the Riesz map, T_g is unitary, so, it's inverse is the Hilbert space adjoint T_g^*. Omitting the names of the bases, $[T_g^*] = [T_g]^*$. The operator adjoint $T_g^{\rtimes}$ with the matrix $[T_g^{\rtimes}] = [T_g]^t$, the transpose. However, the arrow for T_g^1 is not in the original diagram, which we will see later.

The Riesz map P turns a linear form into a vector, while it's inverse τ : V $\rightarrow$ V* maps a vector to a linear form, both are conjugate isomorphisms. This is all we need to show that T_g^1 is unitary.

In the following theorems, we use that V has orthonormal basis B and that V* has the orthonormal basis x.

Proposition 2.1: $\forall$ f $\in$ V*, the coordinates of it's Riesz vector are given by $[P(f)]_e = (\overline{f(e_1)},\overline{f(e_2)},...,\overline{f(e_n)})$.

Proof : Writing τ for the inverse of P, we need to show that
$P(f) = \sum_{i=1}^n \overline{f(e_i)}e_i \Rightarrow$ f $= \tau(\sum_{i=1}^n \overline{f(e_i)}e_i)$.

It is sufficient to show the latter for values of f on the basis vectors e_j, $1 \leq j \leq n$. We obtain

$(\tau(\sum_{i=1}^n \overline{f(e_i)}e_i))(e_j) = \langle e_j, (\sum_{i=1}^n \overline{f(e_i)}e_i)\ \rangle = \sum_{i=1}^n \langle e_j, (\overline{f(e_i)}e_i)\rangle$
$= \overline{\overline{f(e_j)}} \sum_{i=1}^n \langle e_j, e_i\rangle = f(e_j).1$
$= f(e_j)$ for each j.

In particular, this implies that $P(\mathrm{x}_i) = \mathrm{e}_i$.

Proposition 2.2: This inner product on V* satisfies $\langle f, g\rangle = \langle P(f), P(g)\rangle$, where f, $g \in$ V*.

Proof : From definition, $\langle f, g\rangle = [f]_{x,x}.\overline{[g_{x,x}]}$. It is enough to show that $\langle x_i, x_j\rangle = \langle P(x_i), P(x_j)\rangle$. From above,

$\langle P(x_i), P(x_j)\rangle = \langle e_i, e_j\rangle = \delta_{ij} = e_i.e_j = [x_i]_{x,x}.\overline{[x_j]_{x,x}}$.
The result follows.

Hence, one can relate the Riesz vector of $f \in$ V* to the Riesz vector of $f\circ$T$_g^{-1}$. Recall that the Riesz vector of f is the unique vector w $= P(f)$, such that f(v) $= \langle v, w\rangle$ for all v $\in$ V. If $f \neq 0$, it can be found by scaling any non-zero vector in the cokernel of f, which is one-dimensional.

Proposition 2.3: Let T$_g$:V $\rightarrow$ V be unitary, $f \in$ V*, w $= P(f)$ the Riesz vector of f in V*. Then, T$_g$(w) is the Riesz vector of $f\circ$T$_g^{-1}$ ($=$ T$_g^1(f)$).

Proof : Assume $f \neq 0$. Consider the following:

$\langle w\rangle \oplus \ker(f) \longmapsto \langle T_g(w)\rangle \oplus \ker(f\circ T_g^{-1})$, under T_g map. Here, $\oplus$ denotes the orthogonal direct sum.

We need to show that $f\circ T_g^{-1}(v) = \langle v, T_g(w)\rangle$ for all $v \in V$. Since, $w = P(f)$, the Riesz vector of f, we have $f(v) = \langle v,w\rangle$ for all $v \in V$. We obtain $f\circ T_g^{-1}(v) = \langle T_g^{-1}(v),w\rangle = \langle v,T_g(w)\rangle$ as T_g is unitary.

Hence, $T_g(w)$ is the Riesz vector of $f\circ T_g^{-1}$.

The above propsition results in commutativity of the diagrams :

$$\begin{array}{ccc} V^* & \xrightarrow{T_g^1} & V^* \\ {\scriptstyle P}\downarrow & & \downarrow{\scriptstyle P} \\ V & \xrightarrow{T_g} & V \end{array}$$

and

$$\begin{array}{ccc} V^* & \xrightarrow{(T_g^1)^{-1}} & V^* \\ {\scriptstyle P}\downarrow & & \downarrow{\scriptstyle P} \\ V & \xrightarrow{(T_g)^{-1}} & V \end{array}$$

It follows that $P\circ T_g^1 = T_g \circ P$ and $P\circ(T_g^1)^{-1} = (T_g)^{-1} \circ P$.

Proposition 2.4: The first induced operators T_g^1 is unitary.

Proof : We use the fact that T_g is unitary. So, $(T_g)^{-1} = (T_g)^*$ and $\langle T_g(v),w\rangle = \langle v,(T_g)^{-1}(w)\rangle = \langle v,(T_{g^{-1}})(w)\rangle$———————————$(*)$

Let $f, h \in V^*$ be arbitrary, $w = P(f)$ and $u = P(h)$. Then,
$\langle (T_g^1)(f), h\rangle = \langle (P\circ T_g^1)(f), P(h)\rangle = \langle (T_g\circ P)(f), P(h)\rangle = \langle (T_g(P))(f), P(h)\rangle$
$= \langle T_g(w),u\rangle = \langle w,T_g^{-1}(u)\rangle$———(by $*$)
$= \langle P(f),T_g^{-1}(P(h))\rangle$
$= \langle P(f),T_g^{-1} \circ P(h)\rangle$
$= \langle P(f), (P\circ(T_g^1)^{-1})(h)\rangle$
$= \langle P(f), (P((T_g^1)^{-1})(h))\rangle = \langle f,(T_g^1)^{-1}(h)\rangle$

After having looked at eigen values, we see that this generalizes to higher degree that T_g^d is unitarily diagonalizable $\forall$ d $\in \mathbb{Z}^+$. The matrix version of the above theorem is stated as follows.

Proposition 2.5: $[T_g^1]_{x,x} = \overline{[T_g]_{e,e}}$

The Reynold's Operator :

Definition: Define the operators :
$\hat{T}$: V $\to$ V by $\hat{T}$(v) := $\frac{1}{|G|}\sum_{g\in G} T_g$(v) and
$\hat{T}^1$: V* $\to$ V* by $\hat{T}^1$(f) := $\hat{T}^1$(f) := $\frac{1}{|G|}\sum_{g\in G} T_g^1$(f)
which are the Reynold's operators of G.

Proposition 2.6: $\hat{T}$ and $\hat{T}^1$ are self adjoint (Hermitian).

Proof :- To show $\langle\hat{T}$(v),w$\rangle$ = $\langle$v,$\hat{T}$(w)$\rangle$ for arbitrary v, w $\in$ V and g$'$ $\in$ G, we have as follows :
$\langle\hat{T}$(v),w$\rangle$ = $\langle$ $\frac{1}{|G|}\sum_{g\in G}$ T$_g$(v),w$\rangle$
= $\frac{1}{|G|}\sum_{g\in G}\langle$T$_g$(v),w$\rangle$
= $\frac{1}{|G|}\sum_{g\in G}\langle$v,T$_g{}^{-1}(w)\rangle$
= $\frac{1}{|G|}\sum_{g\in G}\langle$v,T$_{g^{-1}}(w)\rangle G$
= $\frac{1}{|G|}\sum_{g'\in G}\langle$v,T$_{g'}(w)\rangle$
=$\langle$v,$\hat{T}$(w)$\rangle$.
The same proof, replacing $\hat{T}$ by $\hat{T}^1$, T$_g$ by T$_g^1$, v by f and w by h gives $\langle\hat{T}^1(f), h\rangle = \langle f, \hat{T}^1(h)\rangle$.

Consequently, $\hat{T}$ and $\hat{T}^1$ are unitarily diagonalizable with real spectrum.

Proposition 2.7: The operators $\hat{T}$ and $\hat{T}^1$ are idempotent i.e.
1. $\hat{T}\circ\hat{T} = \hat{T}$.
2. $\hat{T}^1\circ\hat{T}^1 = \hat{T}^1$.

In particular, the eigen values of both operators are either 0 or 1.

Proof : Let s $\in$ G be fixed. Then,
T$_s\circ\hat{T}$ = T$_s\circ\frac{1}{|G|}\sum_{g\in G}$ T$_g$ = $\frac{1}{|G|}\sum_{g\in G}$ T$_s\circ T_g$ = $\frac{1}{|G|}\sum_{g\in G}$ T$_{sg}$
= $\frac{1}{|G|}\sum_{g'\in G}$ T$_{g'}$ = $\hat{T}$.
From this it follows that
$\hat{T}\circ\hat{T}$ = ($\frac{1}{|G|}\sum_{g\in G}$ T$_g$) $\circ\hat{T}$ = $\frac{1}{|G|}\sum_{g\in G}$ T$_g\circ\hat{T}$ = $\frac{1}{|G|}\sum_{g\in G}\hat{T}$
= $\frac{1}{|G|}.|G|.\hat{T} = \hat{T}$.

We conclude that $\hat{T} \circ (\hat{T}$ - id) $= 0$. Thus, the minimal polynomial of T divides the minimal polynomial $\lambda(\lambda - 1)$, so all the eigen values are contained in {0,1}.

Chapter 3

The Spectrum

We begin this section with a proposition.

Proposition 3.1:
1. If v $\in$ V is an eigen vector of T_g with respect to eigen value λ, then, v is an eigen vector of $T_{g^{-1}}$ corresponding to the eigen value $\frac{1}{\lambda}$.
2. If $f \in V^*$ is an eigen vector of T_g^1 for the eigen value λ, then, f is an eigen vector of $T_{g^{-1}}^1$ for the eigen value $\frac{1}{\lambda}$.
3. If $f \in V^*$ is an eigen vector of T_g^1 for the eigen value λ, then, $P(f) \in V$ is an eigen vector of T_g corresponding to the eigen value $\frac{1}{\lambda}$.
4. If v $\in$ V is an eigen vector of T_g for eigen value λ, then $P^{-1}(\mathrm{v})$ $\in V^*$ is an eigen vector of T_g^1 corresponding to eigen value $\frac{1}{\lambda}$.

Proof : Observe that $g.\mathrm{v} = T_g(\mathrm{v})$ and $g.f = f \circ T_g$.
1. $T_g(\mathrm{v}) = \mathrm{g.v} = \lambda \mathrm{v} \Rightarrow g^{-1}.g.\mathrm{v} = g^{-1}.\lambda \mathrm{v} \Rightarrow g^{-1}.g.\mathrm{v} = \lambda g^{-1}.\mathrm{v}$
   $\Rightarrow \mathrm{v} = \lambda g^{-1}.\mathrm{v} \Rightarrow {T_g}^{-1}(v) = \mathrm{g}^{-1}.\mathrm{v} = \frac{1}{\lambda}\mathrm{v}.$
2. $T_g^1(f) = g.f = \lambda f \Rightarrow g^{-1}.g.f = g^{-1}.\lambda.f \Rightarrow g^{-1}.g.f = \lambda g^{-1}.f$
   $\Rightarrow f = \lambda g^{-1}.\mathrm{f} \Rightarrow T_{g^{-1}}^1(f) = g^{-1}.f = \frac{1}{\lambda}f.$
3. $T_g^1(f) = \lambda f \Rightarrow P(T_g^1(f)) = P(\lambda f) \Rightarrow T_g(P(f)) = P(\lambda f)$
   $\Rightarrow T_g(P(f)) = \frac{1}{\lambda}P(f).$
4. $T_g(\mathrm{v}) = \lambda \mathrm{v} \Rightarrow P^{-1}(T_g(\mathrm{v})) = P^{-1}(\lambda \mathrm{v}) \Rightarrow (T_g^1 \circ P^{-1})(\mathrm{v}) = \frac{1}{\lambda}P^{-1}(\mathrm{v})$
   $\Rightarrow T_g^1(P^{-1}(\mathrm{v})) = \frac{1}{\lambda}P^{-1}(\mathrm{v}).$

It follows that if the union of the spectra is considered over all g $\in$ G, then, the same multiset is obtained irrespective of the choice of T_g or T_g^1.

Continuing after Chapter-1, fix one group element g $\in$ G and compare T_g with T_g^d for d > 1.
Then, by method of stars and bars, we get

$\bar{d} := \dim_{\mathbb{C}}(R_d) = \frac{(n+1+d)!}{(n-1)!d!}$

Every T_g^1 is unitarily diagonalizable with eigen values of absolute value 1. If spec(T_g^1) = $(w_1, w_2, ..., w_n)$, then, V^* has an orthonormal basis, say

$y_g^1 := (y_1, y_2, ..., y_n)$, such that $T_g^1(y_i) = w_i y_i$, $\forall$ 1$\leq$ i $\leq$n and

$[T_g^1]_{y_g^1, y_g^1} = \text{diag}(w_1, w_2, ..., w_n)$

For d > 1, put

$x^d := (x_1^d, x_2^d, ..., x_n^d, x_1^{d-1}x_2, x_1^{d-1}x_3, ..., x_1^{d-1}x_n, ...) =:(\tilde{x_1}, \tilde{x_2}, ..., \tilde{xd})$

all monomials in the x_i of total degree d, numbered from 1 to $\tilde{d}$. These are linearly independent(having no relations among variables) and span R_d forming it's basis. We will not require that this can be made into an orthonormal basis, we do not even consider any inner product on R_d. Rather we establish that

$y^d := (y_1^d, y_2^d, ..., y_n^d, y_1^{d-1}y_2, y_1^{d-1}y_3, ..., y_1^{d-1}y_n, ...) =: (\tilde{y_1}, \tilde{y_2}, ..., \tilde{yd})$ is a basis of eigen vectors of T_g^d diagonalizing it.
Arranging the eigen vectors of T_g^1 similarly,
$(w_1^d, w_2^d, ..., w_n^d, w_1^{d-1}w_2, w_1^{d-1}w_3, ..., w_1^{d-1}w_n, ...) =: (\tilde{w_1}, \tilde{w_2}, ..., \tilde{wd})$.

It follows from above that $T_g^d(\tilde{y_i}) = \tilde{w_i}\tilde{y_i}$, $\forall$ 1 $\leq$ i $\leq$ $\tilde{d}$.

As a consequence, R_d has a basis of eigen vectors of T_g^d and this T_g^d is similar to the diagonal matrix diag$(\tilde{w_1}, \tilde{w_2}, ..., \tilde{wd})$.

For f $\in$ R and g $\in$ G, we say that f is an invariant of g if g.f = f and that f is an invariant of G if $\forall$ g $\in$ G, g.f = f.

Proposition 3.2: For f $\in$ V^*, put $\hat{f} := \hat{T^1}$(f). Then, $\hat{f}$ is an invariant of G.

Proof :- To show that g.$\hat{f} = \hat{f}$ for any arbitrary g $\in$ G, we have

g.$\hat{f} = \hat{f} \circ T_{g^{-1}} = (\hat{T^1}(\text{f}))\circ T_{g^{-1}} = (\frac{1}{|G|}\sum_{s\in G} T_s^1(f)) \circ T_{g^{-1}}$

$\Rightarrow g.\hat{f} = (\frac{1}{|G|}\sum_{s\in G} f \circ T_{s^{-1}}) \circ T_{g^{-1}}$

$= \frac{1}{|G|}\sum_{s\in G} f \circ T_{s^{-1}} \circ T_{g^{-1}}$

$= \frac{1}{|G|}\sum_{t\in G} f \circ T_{t^{-1}} = \hat{f}$

Define the *algebra of invariants* $R^G := \{f \in R : \forall\ g \in G,\ g \circ f = f\}$. This forms a subalgebra of R. The grading in R^G is similar to that in R.

Proposition 3.3: The algebra of invariants of G is naturally graded as follows :
$R^G = \oplus_{d \in \mathbb{N}} R_d^G$, where $R_d^G = \{\ f \in R_d : \forall\ g \in G,\ g \circ f = f\ \}$ is the d-th homogeneous component of R^G.

Proof : This follows directly from propositions 1.1 and 1.2.

Chapter 4

The Series and the theorem

Considering R_d^G as a vector space, define $a_d := \dim_{\mathbb{C}} R_d^G$, the number of linearly independent homogeneous invariants of degree d $\in \mathbb{N}$
Then, $\phi_G(\lambda) := \sum_{d\in\mathbb{N}} a_d \lambda^d$ is the *Molien series* of G, whose coefficients are the number of linearly independent homogeneous invariants of degree d.

Statement of Molien's theorem : If $\phi_G(\lambda)$ defined as above is the generating function for this sequence, then, $\phi_G(\lambda) := \frac{1}{|G|}\sum_{g\in G}\frac{1}{det(id-\lambda T_g)}$

Proof :- First, we show that $a_1 = \mathrm{Tr}(\hat{T}) = \mathrm{Tr}(\hat{T}^1)$.
Now, $\mathrm{Tr}(\hat{T}) = \mathrm{Tr}(\hat{T}^1)$ follows from the remark after Proposition 3.1 since the sum for the trace runs over all group elements. From Proposition 2.7, both the operators are idempotent, Hermitian and V^* has an orthonormal basis $f = (f_1, f_2, ..., f_n)$ of eigen vectors of $\hat{T}^1$ corresponding to the eigen values $\lambda_1, \lambda_2, ..., \lambda_n \in \{0,1\}$. So, $[\hat{T}^1]_{f,f} = \mathrm{diag}(\lambda_1, \lambda_2, ..., \lambda_n)$.

Let us assume that this matrix has the first r entries 1 and remaining n-r entries 0(possible by rearranging)

i.e. $[\hat{T}^1]_{f,f} = \begin{cases} 1 & : 1 \le i \le r \\ 0 & : r+1 \le i \le n \end{cases}$

Hence, $\hat{T}^1(f_i) = \begin{cases} \mathrm{f}_i & : 1 \le i \le r \\ 0 & : r+1 \le i \le n \end{cases}$

Any linear invariant of G is certainly fixed by $\hat{T}^1$, so $a_1 \le$ r. On the other hand, by Proposition 3.1, $\hat{f}_i = \hat{T}^1(f_i) = \lambda_i f_i$ is an invariant of G for every $1 \le$ i $\le$ r, so $a_1 \ge$ r. Hence, $a_1 =$ r. So, we have $a_1 = \mathrm{Tr}(\hat{T}) = \mathrm{Tr}(\hat{T}^1)$ and $a_d = \mathrm{Tr}(\hat{T}^d)$.
Again, by definition, $\hat{T}^d = \frac{1}{|G|}\sum_{g\in G} T_g^d$.

Now, denote by $[\lambda^i]$: the coefficient of λ^i in p(λ) $\in \mathbb{C}[\lambda]$.
Then, $a_d = \frac{1}{|G|}\sum_{g\in|G|}\mathrm{Tr}(T_g^d)$ and thus, all we need to show is
$[\lambda^d] : \frac{1}{|G|}\sum_{g\in G}\frac{1}{det(id-\lambda T_g^1)} = \frac{1}{|G|}\sum_{g\in G}\mathrm{Tr}(T_g^d)$.
We show that for every summand(group element), the equation:
$[\lambda^d] : \frac{1}{det(id-\lambda T_g^1)} = \mathrm{Tr}(T_g^d)$ holds.
$\forall$ g $\in$ G, $\mathrm{Tr}(T_g^d) = \mathrm{Tr}(\mathrm{diag}(\tilde{w_1}, \tilde{w_2}, ..., \tilde{wd}))$
$= \tilde{w_1} + \tilde{w_2} +...+ \tilde{wd}$
= sum of the products of $\tilde{w_1}$, $\tilde{w_2}$,..., $\tilde{w_n}$ taken $\tilde{d}$ of them at a time.
For the same g $\in$ G, one obtains that $[T_g^1]_{y_g^1, y_g^1} = \mathrm{diag}(\tilde{w_1}, \tilde{w_2}, ..., \tilde{w_n})$ so that
$\det(id - \lambda T_g^1) = \det(id - \lambda.\ \mathrm{diag}(\tilde{w_1}, \tilde{w_2}, ..., \tilde{w_n})\ )$

$\Rightarrow \det(id - \lambda T_g^1) = (1-\lambda w_1)(1-\lambda w_2)...(1-\lambda w_n)$
$\Rightarrow \frac{1}{det(id-\lambda T_g^1)} = \frac{1}{(1-\lambda w_1)(1-\lambda w_2)...(1-\lambda w_n)}$
$\Rightarrow \frac{1}{det(id-\lambda T_g^1)} = (1-\lambda {w_1}^{-1})(1-\lambda {w_2}^{-1})...(1-\lambda {w_n}^{-1})$
$\Rightarrow \frac{1}{det(id-\lambda T_g^1)} = (1+\lambda w_1+\lambda^2 w_1^2+...)(1+\lambda w_2+\lambda^2 w_2^2+...)...(1+\lambda w_n+\lambda^2 w_n^2+...)$
and here, the coefficients of λ^d is also sum of the products of $w_1, w_2, ..., w_n$ taken d of them at a time.
As the sum runs over all group elements, it follows that
$\frac{1}{|G|}\sum_{g\in G}\frac{1}{det(id-\lambda T_g)} = \frac{1}{|G|}\sum_{g\in G}\frac{1}{det(id-\lambda T_g^1)}$.
The result follows.

An Example :

Consider the symmetric group S_3 acting on $\mathbb{R}^3$ by permuting the coordinates. Add up the sum of group elements as follows.
Starting with identity,

$$\begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{vmatrix} = (1-\lambda)^3.$$

There is a three element conjugacy class of S_3 consisting of the swap of two coordinates. This gives three terms of the form :

$$\begin{vmatrix} 1 & -\lambda & 0 \\ -\lambda & 1 & 0 \\ 0 & 0 & 1-\lambda \end{vmatrix} = (1-\lambda)(1-\lambda^2).$$

There is a two element conjugacy class of the cyclic permutations yielding two terms of the form :

$$\begin{vmatrix} 1 & -\lambda & 0 \\ -\lambda & 1 & 0 \\ -\lambda & 0 & 1 \end{vmatrix} = (1-\lambda^3).$$

Notice that different elements of the same conjugacy class yield the same determinant.

Assembling these results give
$\frac{1}{|S_3|}\sum_{g\in S_3}\frac{1}{det(id-\lambda P(g))} = \frac{1}{6}(\frac{1}{(1-\lambda)^3} + \frac{3}{(1-\lambda)(1-\lambda^2)} + \frac{2}{1-\lambda^3})$
$\Rightarrow \phi_{S_3}(\lambda) = \frac{1}{(1-\lambda)(1-\lambda^2)(1-\lambda^3)}$
i.e. the ring of invariants is generated by three elements, a linear invariant, a degree 2 invariant and a degree 3 invariant.
Expanding the terms in the generating function,
$\phi_{S_3}(\lambda) = (1+\lambda+\lambda^2+...)(1+\lambda^2+\lambda^4+...)(1+\lambda^3+\lambda^6+...)$
$\Rightarrow \phi_{S_3}(\lambda) = (1+\lambda+2\lambda^2+3\lambda^3+...)$
The coefficients of this series gives the number of linearly independent homogeneous polynomials in three variables, which are invariant under permutations of three varaibles.

Chapter 5

Conclusion

There is a version of *Molien's series*, which gives the notion of the homogeneous components $K[V]_d$ in modular case (i.e. when the characteristic of K divides the order of the group $|G|$). The computations in Molien's formula are dependent only on the conjugacy class and holds only for non-modular case.

For a finite dimensional KG-module V, a modification of *Molien's series* yields the extended Hilbert's series of the module of invariants if the Brauer character table of G is known. *Molien's series* provides an efficient algorithm to find the generating invariants of some large groups, for example, the rotation group of a cube in $\mathbb{R}^3$.

Some places to explore :

- If we know about the conjugacy classes of G, we may be able to say more, since every unitary representation splits into irreducible components.
- There seems to be a link with Polya's enumeration.
- Relations on the generators in S of the Cayley graph Γ(G,S) should lead to conditions of the minimal polynomial of it's adjacency Q(Γ(G,S)).

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